

**DEPARTMENT OF CHEMICAL & BIOMOLECULAR ENGINEERING
NORTH CAROLINA STATE UNIVERSITY**

CHE 330

Homework Set 5

Note: this assignment is submitted via TurnItIn. If you do not have a TurnItIn account, use the instructions provided on the CHE 330 Moodle site to create an account.

Though experimental design principles are typically used to evaluate and improve industrial processes, the same principles can also be applied to phenomena encountered in daily life. This homework assignment is to complete and evaluate a 2-level full factorial experimental design with 3 independent variables on the topic of your choice. Examples of suitable studies include:

1. Hike

- o Independent Variables (Factors)
 - Legwear (pants, shorts)
 - Shoes (hiking shoes, running shoes)
 - Size of meal prior to hike (200 calories, 800 calories)
- o Dependent Variable (Response)
 - Time to complete hike (minutes)

2. Beverage Chilling

- o Independent Variables (Factors)
 - Container materials (can, bottle)
 - Cooling method (refrigerator, freezer)
 - Time (20 minutes, 40 minutes)
- o Dependent Variable (Response)
 - Temperature (degrees Fahrenheit)

3. Popcorn

- o Independent Variables (Factors)
 - brand of popcorn (ordinary, gourmet)
 - amount of dry popcorn (1/3 cup, 2/3 cup)
 - popcorn-to-oil ratio (low, high)
- o Dependent Variable (Response)
 - Volumetric yield of popcorn (cups)

You may choose to test any topic and factors you wish. However, the tested system must have an objective, quantitative response – subjective measures such as taste are not valid. You do not need to show your calculations for factor and interaction effects, but your raw data is required. Summarize your experiment and responses to Problems 1-3 below in a brief report (i.e. introduction, body and conclusion – use a format of single or double-spaced 12 pt Times New Roman font, 2 pages maximum):

Problem 1 (60%) Execute a 2^3 full factorial experimental design for your study and describe findings via a table. Determine the effect (θ) for each factor and any interactions. Which factors and interactions contribute most/least to system response? What are the practical implications?

Problem 2 (30%) Develop a predictive model equation for the studied system. Using the model, determine a predicted response for untested values of one or more factors. Then, perform a trial using these previously-untested values and compare experimental response to model prediction. Did the model accurately predict system response? What limitations may have impacted the prediction?

Problem 3 (10%) What did you learn about the studied system that you didn't know before?